

Software Requirements Specification (SRS)

Matrix Inquiry Module

Document Control

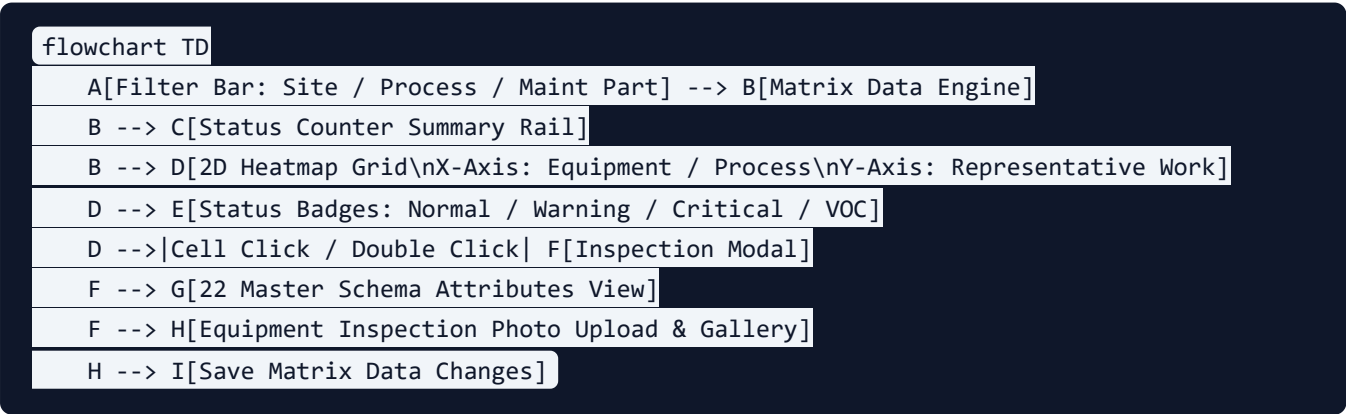
Item	Details
Document Title	SRS: Matrix Inquiry & Equipment Heatmap Module
System Module	EQUAL Change Matrix Subsystem (/change-matrix/matrix-inquiry)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Data Visualization Engineers, Maintenance Managers, QC Test Engineers

1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **Matrix Inquiry Module** provides an interactive 2D Heatmap Grid cross-referencing equipment assets against representative work activities, enabling plant engineers to visualize maintenance frequency, operational status, and VOC (Voice of Customer) complaint flags.

1.2 Individual Module Flow Diagram



2. Functional Requirements

2.1 2D Heatmap Matrix Grid

- **Req-1.1 Axis Alignment:** X-Axis represents Equipment Items / Process Steps; Y-Axis represents Representative Work Items.
- **Req-1.2 Status Color-Coding:**
 - Normal : Green badge.
 - Warning / Frequent : Amber badge.
 - Critical / VOC Flagged : Red badge.

2.2 Inspection Modal & Media Attachments

- **Req-2.1 Detail Inspection:** Clicking a matrix cell opens a modal displaying detailed work order records across the 22 master schema fields.
- **Req-2.2 Photo Attachments:** Supports uploading, viewing, and managing equipment inspection photos.

3. QC Testing & Acceptance Criteria

- **Filter Synchronization:** Grid MUST reload dynamically when Site or Process filter selections change.
- **Heatmap Color Accuracy:** Status color badges MUST accurately reflect equipment criticality and VOC status.
- **Photo Upload Verification:** Attached inspection photos MUST render correctly within the cell detail modal.